# A role-symmetric logic and dependent calculus: research assessment of K-FOG

Research notes for the KFOL workspace

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**Assessment.** The central object is mathematically viable, and there is a particularly direct precedent: Abramsky's $L_A$. The role-indexed quantifier idea, including independent per-agent valuations, should not be claimed as new. There are also existing multirole proof calculi and role-indexed dependent session quantifiers. These findings support the idea's coherence, while substantially changing its novelty position. I have not established that the desired full dependent proof calculus already exists, nor established a novel theorem for it.

The object of interest here is intrinsically multi-role interaction, rather than a language organized around individual forceability. The latter is one optional observation of a game and is not used to define the proposed core. Several forceability claims are audited in the appendix because the manuscript makes them. The original manuscript has not been edited.

## 1 Which existing objects actually match?

**The closest semantic match: Abramsky's $L_A$.** A Compositional Game Semantics for Multi-Agent Logics of Partial Information (2007), especially Sections 2.4, 3.3–3.4, and 4.4–4.6, gives role-owned quantifiers, role permutations, strategy interaction, per-agent valuations, and partial information. It treats formulas as interaction structures. Its Section 6 leaves proof theory open; that is a statement about that paper, not a claim that the problem remains open today.

For the manuscript's broadcast prenex fragment the comparison can be made exact, using its intended full-history strategies rather than the inconsistent free-variable wording in its strategy definition. Translate each $P_i x$ to a choice particle owned by $i$. Package the terminal $\langle \varphi_0, \ldots, \varphi_{K-1} \rangle$ as an interpreted terminal atom $a_{\vec{\varphi}}(\vec{x})$ whose $i$-valuation is

$$v_i(a_{\vec{\varphi}}, \mathcal{M}, s) := [\mathcal{M}, s \models \varphi_i].$$

This is a definitional interpretation of a new terminal symbol, not a claim that every tuple is already an atom in the original first-order signature. It gives the candidate correspondence

$$v_i\big(\mathsf{play}(T\Phi, T\sigma)\big) = [\![\Phi]\!]_{\mathcal{M},s,\sigma,i}. \tag{1}$$

For a fixed broadcast prefix, strategies in both presentations can be restricted to the same history-to-choice functions. An induction on the prefix then proves (1): the same owner selects the same value, and the terminal clause is true by definition. This is a correspondence of full evaluations, before choosing a success predicate. For arbitrary information annotations, one must additionally prove that the two presentations admit exactly the same observation-uniform strategies. The manuscript's inconsistent slash conventions cannot be translated as a single convention without repair.

**A close algebraic antecedent.** Olde Loohuis and Venema, Logics and Algebras for Multiple Players (2010) uses player-owned connectives, role switches, and valuations declaring arbitrary subsets of players winners. It develops equational proof systems and representation results. Its chosen equivalence observes winning-strategy sets; retaining the complete interaction object requires a finer equivalence. This is a direct antecedent for the multi-outcome idea, even though it is not the desired dependent term calculus.

**A close proof-theoretic antecedent.** Xi and Wu, Multirole Logic and Multiparty Channels, with earlier work in 2017 and 2016, develops role-indexed logical operations and multiparty cut elimination. Its first-order quantifiers are indexed by ultrafilters; on a finite role set these are principal. It includes classical, intuitionistic, and linear variants. This is evidence for a real proof theory of role-indexed logic, not merely a protocol analogy. It does not automatically validate the manuscript's unrestricted payoff tuples or proposed compositions.

**The dependent quantifier already has an implementation precedent.** Wu and Xi, Multiparty Dependent Session Types (2018), Figure 7 and Section 4, has $\mathsf{quan}(r, \lambda a : \sigma.\pi)$. Its interpretation varies by endpoint, with the designated role interpreted universally and the others existentially in that presentation. The polarity convention differs from an owner-as-witness presentation; the neutral role-indexed mechanism is already there. The calculus separates statics and dynamics, so this result alone is not an identification with a full CIC-style dependent calculus.

**Other comparisons a type-theory claim must survive.** There are existing game models of dependent type theory, including Yamada's work and Abramsky–Jagadeesan–Vákár. Those results show that dependent game semantics itself is not new. They do not by themselves supply this particular multi-role object language. Escardó and Oliva model dependent game trees and mechanize strategy constructions in Agda. Using dependent types to describe games is therefore also an existing direction. Open games supply a compositional treatment of game-theoretic interaction and equilibria; a composition claim should compare the preserved behavior, not merely the use of sequential and parallel notation.

For the dependent information discipline, compare A Dependent Dependency Calculus and Multimodal Dependent Type Theory. The former integrates dependency analysis into dependent typing; the latter gives a general framework for interacting modalities with consistency and canonicity results. Neither is automatically an identification with neutral multi-party choice. The current Kami theory repository explicitly describes a distributed language based on MTT with full dependent types, so a broad claim about being the first dependent multi-location calculus would be particularly unsafe.

## 2 The object to retain

A useful semantic object is

$$G = \big(\text{typed interaction structure}, (\mathsf{Str}_i(G))_{i<K}, \mathsf{play}_G, (B_i)_{i<K}\big). \tag{2}$$

Here $\mathsf{play}_G$ resolves a profile of strategies into a complete history, and $B_i$ is the $i$-th terminal meaning on that history. In the manuscript $B_i$ is Boolean-valued. For a dependent theory it can instead be a family of propositions or evidence types. General objectives remain general: no one-hot or common-goal condition is imposed on (2).

No strategy is selected or declared successful in the definition of (2). One can subsequently ask about proofs, partial implementations, refinement, equilibria, or forceability. Keeping these separate prevents the latter from silently becoming the whole meaning of a formula.

For a fixed prefix, number occurrences $m = 1, \ldots, n$ and attach an owner $r_m$, an action type, and an observation map

$$o_m : H_m \longrightarrow O_m.$$

For fixed nondependent action types $A_m$, pure strategies are exactly

$$\mathsf{Str}_i(G) = \prod_{m : r_m = i} (O_m \longrightarrow A_m). \tag{3}$$

This gives a precise meaning to a single strategy for a role: a family of local policies, one per occurrence. Additional coherence requirements must be stated as additional requirements; packaging the policies into one function does not create them.

In a dependent version, start with action types that descend to observations: there is a family $\bar{A}_m : O_m \to \mathcal{U}$ and

$$A_m(h) = \bar{A}_m(o_m h), \qquad f_m : \prod_{o : O_m} \bar{A}_m(o). \tag{4}$$

This avoids assigning a player a choice type that varies with an unavailable secret. A richer theory could allow abstract hidden witnesses, but would need explicit packaging or a modal discipline rather than an unstated coercion.

When observations are sets of earlier columns $K_m$, perfect recall has a simple checkable sufficient condition: whenever $b < m$ have the same owner,

$$K_b \cup \{b\} \subseteq K_m. \tag{5}$$

The unrestricted occurrence-local calculus and the perfect-recall calculus are both legitimate objects. They are different objects. The manuscript currently uses properties of both without fixing which one is intended.

## 3 Three precise foundational facts

**Proposition 1** (Evaluation and change of schedule)**.** *For an acyclic finite dependency structure, total deterministic local policies of the form (3) determine a unique complete history. Changing to another topological order of the same dependency structure does not change that history.*

*Proof.* For any event, its predecessors determine its observation and hence its action. Induct over the dependency order. Each event receives the same inputs in every topological execution, so it produces the same value. This argument uses the same event identities, action types, and policies; merely rearranging a broadcast prefix without preserving observations does not meet the hypothesis. □

**Proposition 2** (Role equivariance)**.** *A permutation of roles transports (2), its strategies, and its complete evaluations equivariantly.*

*Proof.* Relabel owners, reorder the terminal family, and transport each occurrence's policy along the corresponding occurrence bijection. The values selected at corresponding events are equal by induction. Thus the resulting terminal vector is permuted. If policies take syntax as an argument, the transport must also rename that argument; reordering the profile coordinates alone is insufficient. □

**Proposition 3** (No single-role permutation is general duality)**.** *For $K > 2$, no permutation of the $K$ individual roles implements Boolean complement on every one-hot terminal outcome.*

*Proof.* A permutation takes a one-hot vector to a one-hot vector. Its Boolean complement has $K - 1$ nonzero coordinates. These cannot be equal when $K > 2$. □

The last fact explains a structural distinction. For local producer/consumer behavior, if an endpoint owns a set $R$ of roles, the complementary behavior at every role-owned choice belongs to the complementary set $\bar{R}$. The complement of a singleton is not a singleton when $K > 2$. Multiparty interaction therefore needs a coherent family of endpoints, often indexed by a partition of roles. This does *not* justify defining logical negation by complementing role sets: the interaction laws, identity, and cut must still be checked. The multirole literature itself identifies failures of naive negation under multiparty cut.

# 4 A small dependent kernel worth making exact

Take a terminating dependent data calculus as a base. Initially all communicated data and their types are public. Use a neutral global constructor

$$\mathsf{Q}_r(x : A).G$$

with the continuation $G$ allowed to depend on the chosen value. Role labels remain part of the object language. A local interpretation at role $i$ can be defined, using the owner-as-witness convention, by

$$L_i(\mathsf{Q}_r(x : A).G) = \begin{cases} \sum_{x:A} L_i(G), & i = r, \\ \prod_{x:A} L_i(G), & i \neq r. \end{cases} \tag{6}$$

This is a concrete constructive formulation of local witness production and local input consumption. It is not a claim of novelty: it is close to the dependent session quantifier mechanism already discussed. For ordinary behavior types set $L_i(\mathsf{End}) = \mathbf{1}$. For certified endpoint types, optionally use an evidence family $L_i(\mathsf{End}) = B_i$. The global object and its ordinary behaviors exist independently of whether any certified endpoints exist.

For example, with inhabited $X$ and positive natural numbers $\mathbb{N}_{>0}$,

$$\mathsf{Q}_A(n : \mathbb{N}_{>0}).\ \mathsf{Q}_B(v : \mathsf{Vec}(X, n)).\ \mathsf{Q}_C(j : \mathsf{Fin}(n)).\ \mathsf{End}(B_A(n, v, j), B_B(n, v, j), B_C(n, v, j)) \tag{7}$$

contains an actual dependency of later choice types on earlier choices. At $A$ the first constructor is a dependent pair; at $B$ and $C$ it is a dependent function. Subsequent constructors make different roles the producer. The tuple of terminal families is retained globally.

The interaction reduction at the head choice is explicit:

$$\mathsf{run}\big(\langle a, t_r\rangle, (\lambda x.t_i)_{i\neq r}\big) \longrightarrow \mathsf{run}\big(t_r, (t_i[a/x])_{i\neq r}\big). \tag{8}$$

More generally apply arbitrary dependent-function terms to $a$; lambdas merely display the beta step. The chosen value is substituted in all continuation types as well as their terms.

**Proposition 4** (Dependent interaction for a fixed finite telescope)**.** *Given local inhabitants of all the types (6), interaction produces one common history. If terminal types are $B_i$, it also produces an inhabitant of each $B_i$ at that same history. There is one head communication per global binder.*

*Proof.* At a head owned by $r$, eliminate the dependent pair at endpoint $r$, obtaining $a : A$ and its continuation. Apply every other endpoint's dependent function to $a$. The results have exactly the local types of the continuation with $a$ substituted. Recurse on the shorter telescope. At $\mathsf{End}$ the endpoint terms already inhabit the terminal families. Termination of the communications follows from the decreasing telescope length; normalization of arbitrary data-level computations is a separate property of the base calculus. □

This is already stronger than a Gallina-looking let-chain as an account of typed computation. But it does not yet give implication between interacting objects, higher-order interaction, a proof theory for all of (2), or a new foundation for dependent mathematics. Those are separate claims.

## 5 What would be a substantial research contribution?

The most useful candidate is a *dependent multiparty substitution/cut theorem for the full interaction object*. It would give typed open terms, interfaces, identity, application/composition, and a reduction semantics whose meaning preserves role ownership and dependency. It should retain arbitrary terminal families and the behavior of each role, rather than prove a theorem only about one Boolean projection.

Here are concrete proof obligations, in an order that avoids claiming more than has been built:

1. **Correspondence to the existing semantic object.** Formalize (1), including the admissible strategy families. This settles exactly which part of the manuscript is an existing fragment and which is an extension.

2. **An actual dependent term calculus.** Specify formation, introduction, elimination, substitution, and judgmental equality for a small neutral choice fragment. Prove (8) preserves typing and the appropriate observations. Merely defining a datatype of games inside CIC does not establish a new object type theory.

3. **Identity and implication.** Define what a term using $G$ to implement $H$ is, with its role-indexed interface. The manuscript's material implication between terminal goals is not such a definition: it supplies no application or substitution operation. State the required coherence at shared terminals instead of importing an identity axiom for arbitrary goal tuples.

4. **Composition of open terms.** Prove substitution is associative up to the declared equality and gives identity laws. For an information-aware extension, require an interface substitution to respect observation equivalences. In elementary terms, if $h \sim_i h'$, then the substituted input views available to $i$ must still agree. This condition composes and preserves uniform local policies. Extending it to dependent types is a substantive part of the design.

5. **Computation and checking.** Prove normalization or an appropriate weaker result for the chosen fragment, and establish decidable checking under explicit assumptions on the base calculus. Cut elimination alone does not imply confluence. Classical proof reduction has well-known confluence difficulties; see Hetzl and Straßburger. Ownership may help orient interactions, but is not a proof that every higher-order critical pair is joinable.

6. **A precise conservativity claim.** Give a translation of an ordinary dependent fragment and prove the claimed preservation and reflection. Greater expressiveness about interaction does not automatically mean greater proof-theoretic strength or a stronger consistency assumption.

The first two items are reliable starting points and are not, on their own, compelling novelty claims. The remaining items are candidate research results. I have not proved them for a specified new calculus or established that no existing system proves an equivalent theorem. Comparison must be against the multirole systems, dependent session systems, and modal/dependency calculi above.

One subtlety in the existing intuitionistic multirole presentation is that its intuitionistic judgment is parameterized by a fixed ultrafilter. Over finite roles that parameter selects a principal role. An equivariant family of such judgments is different from one fixed judgment with a privileged role. Whether the desired theory needs a single fully neutral judgment, or an explicitly role-indexed family of constructive judgments, is an exact design issue. Neither choice should be concealed by notation.

I would initially keep universes predicative and interaction finite, with dependent data and a small genuine composition fragment. General recursion, impredicativity, full inductive families of interactive types, and arbitrary information disclosure can be separate extensions. A small proof of a meaningful cut theorem would settle more than a large collection of encodings or complexity reductions.

# A Manuscript audit

The following audit distinguishes false claims from repairable definitions and results that are essentially correct under explicit hypotheses. File locations refer to the unchanged manuscript in this workspace.

## A.1 Claims that survive

- Evaluation by recursion on a finite prefix is well-defined for total strategies and nonempty action domains. If the domain is infinite, the game is finite-horizon, not necessarily a finite game tree.
- Role equivariance is correct after transporting syntax-indexed strategies.
- The two-player, complementary-payoff prenex embedding has the expected projection and role-swap laws in the broadcast fragment, after fixing the seat-indexing error described below.
- The broadcast individual-projection decision problem is PSPACE-complete already over a fixed Boolean structure. This is a model-checking result, not a decidability claim about validity over all structures.
- Joining total functional strategy graphs on an acyclic dependency structure computes the unique play. This retains counterfactual strategies only if the full graphs are retained, not just their resulting row.
- The tensor/leaky product forceability equality for disjoint player blocks, unchanged component objectives, total pure strategies, and broadcast component games is repairable. To extract a strategy on the right, freeze one whole left-hand history consistently. The proof's claim that arbitrary history-by-history extensions can be chosen is too strong. Its first direction also needs attention to the enlarged opponent strategy sets.

These are useful facts but do not make individual forceability the intended core.

## A.2 The abstraction theorem is incorrectly indexed and overclaims uniqueness

In `2FOL.tex:29`, $\mathsf{A}_i(\theta)$ always has terminal $\langle \theta, \neg\theta \rangle$. For $i = 1$ its $i$-projection is $\neg\theta$, contradicting the stated retract. Put $\theta$ in seat $i$ and $\neg\theta$ in the other seat.

The broad uniqueness assertion in `appendix-proofs.tex` does not follow from projection laws. Semantically equivalent redundant structure can be inserted. Moreover, the appendix uses Boolean game constructors absent from the core grammar. Pointwise conjunction of complementary terminal pairs gives $\langle a \wedge b, \neg a \wedge \neg b \rangle$, whose second component is not $\neg(a \wedge b)$. The proof assumes the connective compatibility it needs to establish. A uniqueness statement restricted to a stipulated constructor-by- constructor translation is a different, much narrower statement.

### A.3 Three incompatible information conventions

`info.tex` starts with roles hidden from a newly produced value, then defines independence from earlier variables at the current choice. `session.tex` uses the outgoing recipient convention; `complexity.tex` uses incoming variable independence. These are distinct annotations. Fix one observation-map semantics and translate any surface sugar into it explicitly.

`KFOL.tex:57` describes strategies using assignments only to the free variables of the remaining formula. That can omit earlier observations which are unused by terminal predicates but usable as signals. Use complete histories and then project to the declared observations.

### A.4 The DQBF reduction is false as written

The reduction in `complexity.tex:55` hides only the disallowed universal variables. Earlier existential choices remain visible and can carry forbidden information. Consider the false DQBF

$$\forall y; \quad x_1(y),\ x_2(); \quad (x_1 = y) \wedge (x_2 = y).$$

The manuscript's encoding permits $x_1 := y$ followed by $x_2 := x_1$, so the encoded player wins. The intended $x_2$ must be constant and cannot always equal $y$. For occurrence-local information, repair the hidden set to

$$H_j = (\{y_1, \ldots, y_n\} \setminus D_j) \cup \{x_1, \ldots, x_{j-1}\}. \tag{9}$$

This repair permits forgetting. It does not prove the theorem under strict single-player perfect recall.

For the corrected Boolean finite-prefix calculus with arbitrary occurrence-local views, individual forceability is in NEXPTIME as well as hard for it: guess the exponential-size local truth tables and enumerate all opponent action sequences. A losing sequence is realizable by opponent policies constant at each occurrence, so universal opponent-strategy tables need not themselves be enumerated. This upper-bound argument assumes unrestricted nonempty Boolean moves and a polynomial-size Boolean terminal formula.

With perfect recall for the one forcing role and fixed incoming observation masks, its observed opponent variables form nested sets. Earlier own choices can be inlined as functions of the remembered observations. Place each universal input immediately before the first forcing move that observes it, and place never-observed inputs at the end. This yields an ordinary QBF of polynomial size, giving a PSPACE bound for that restricted problem. This is compatible with the known collapse of perfect-recall IF truth conditions to first-order truth conditions; see the primary Mann–Sandu–Sevenster book preview, discussion of Chapter 6. It does not equate full games, all players' perspectives, or distributed coalitions with a single scalar formula.

### A.5 Nash and individual rationality

`solutions.tex:134` says IR-NE may fail to exist when pure Nash equilibria exist and every player can force a win. Under the given definitions:

$$\mathsf{NE}(\sigma) \quad \Longrightarrow \quad \mathsf{IR}(\sigma).$$

If $i$ could force 1 but received 0, switching to that forcing strategy would be a profitable unilateral deviation. Hence IR-NE and NE are the same set here. Further, if every $i$ has a forcing strategy, choose one for each role. Their joint profile wins for everyone, because each guarantee applies against all others. That profile is automatically Nash for Boolean payoffs. This contradicts the accompanying discussion of independently forcing profiles.

The definitions of pure NE and weak dominance themselves are standard and usable. For finite perfect-information games with nonempty finite action sets, pure subgame-perfect equilibria already exist by backward induction; hidden-information extensions can lack pure NE.

### A.6 Equal objectives and negation

Identical terminal objectives do not turn individual adversarial quantification into collective existence. Over the Boolean domain,

$$P_0 x.\langle x = 0, x = 0\rangle$$

has individual forceability $(1, 0)$ and is jointly realizable. Complementing both goals still gives individual forceability $(1, 0)$, by choosing $x = 1$. Thus the cooperative-negation remark in `KFOL.tex:253` is false.

The general no-De-Morgan conclusion is correct, but its displayed recursive calculation incorrectly replaces the projection of a complemented arbitrary subgame by the negation of its projection. Keep the complemented subgame inside the projection, or restrict the example to a terminal.

The claim of a unique nontrivial player-symmetric involution is valid if one requires the same unary Boolean map independently on every coordinate. It is false for general equivariant involutions: swapping only $0^K$ and $1^K$ while fixing all other vectors is another such involution for $K \geq 2$.

### A.7 Terminal syntax is not a safe semantic quotient

The cheap-talk quotient in `solutions.tex:220` need not preserve outcomes. Let $P_0 c.P_1 y.\langle y = 0, \top\rangle$ and fix 1's policy $y := c$. Changing only $c$ changes the outcome, although $c$ occurs in no terminal goal. A quotient needs an explicitly proven observational congruence.

The slicing lemma in `solutions.tex:495` is false because it switches a whole role's strategy from the irrelevance of just one occurrence. Let $P_1 v.P_1 u.\langle v = 0, \top\rangle$. The final $u$-occurrence is irrelevant, but replacing player 1's full strategy can change $v$. Repair the conclusion to a local-policy change, or require all of that role's occurrences to be irrelevant.

The definition of game equivalence by visibility-DAG isomorphism must include the terminal interpretations and typed labels. Identical graphs with different terminal goals are not equivalent games. Likewise, consistency of every Boolean combination of terminal goals does not establish strategic noninterference.

### A.8 Composition needs more than acyclic graph gluing

The splice in `algebra.tex:704` specifies an arena but no complete terminal composition. Two independently produced copies of a shared boundary value can also disagree: each component may determine a valid singleton play while their join is empty. A cut interface must identify which side produces each value and which side consumes it, or require a compatibility proof.

The suffix/prefix edge conditions in that section are reversed relative to the earlier-to-later orientation. Several clock and megaphone definitions are also incomplete: the clock written before the game is described as after it; a new role lacks a terminal component; fixed echo behavior is an extra operational primitive rather than an ordinary unrestricted role choice. These constructions need definitions before their algebraic laws can be assessed.

Role merging preserves decentralized strategies only when occurrence-level views remain unchanged. Recomputing a merged role's memory or pooling its observations can change the game. The phrase "one coalition acts as a single role" must say which operation is intended.

### A.9 Relational representation and counterfactuals

Totality only on histories reached under a fixed complete profile, as in `ra.tex:101`, is sufficient to run that profile. It is insufficient for deviations. Strategies must be defined on every legal observation that can arise against alternative opponents, not just the equilibrium path.

Excluding a hidden column from a local key prevents direct access; it does not prevent indirect signaling through a visible intermediary. The DQBF countermodel is also a counterexample to a general no-signaling conclusion from key exclusion. Retaining all local tables, including off-path rows, is essential for the game-theoretic reading of relational queries.

### A.10 Public properties and session translations

Exact identifiability in `extensions.tex:43` misses partial information. Over four possible secret values, observing parity can inform a later choice without identifying the secret. Use observation partitions/maps rather than an all-or-nothing visibility edge. A claimed public property also needs evidence or an explicit trust/ideal-functionality assumption. An inconsistent public theory identifies everything. Unique existence alone does not introduce an $\iota$-term into the first-order signature. With all certificates trivial, the proposed private-value extension does not recover the broadcast base, so its stated conservativity claim is not established.

The session translations are at most structural encodings without further conditions. Ground message sorts do not force repeated sends to carry the same value, atomic broadcast knowledge is not supplied by a sequence of ordinary sends, and a private choice can vanish from the target. Refinements checkable in the global context need not be checkable in the sender's local context. Demanding all players' goals as refinements also changes a multi-outcome game into a joint-admissibility problem. The conjecture at `session.tex:137` needs a specified equivalence and a corrected source language before it is a theorem candidate. It is not evidence for a Curry–Howard correspondence as written.

## B Finite checks and their limits

`research/finite_audit.py` enumerates total Boolean policies, retaining off-path behavior and replacing all policies owned by a role during deviations. It confirms the DQBF, hidden-matching, cooperative-negation, cheap-talk, and slicing examples above. It also checks all 256 payoff tables for two Boolean moves in each of two observation regimes: 512 games and 3,072 profiles, with 512 broadcast projection comparisons. All checks passed.

These are finite experiments, not Lean/Rocq proofs. The manuscript's `interpreter.v` is pseudocode with an unfinished assignment definition and does not mechanize its theorems. A useful first formalization would implement the exact game object, role transport, and the finite dependent interaction theorem; the manuscript's full claim set should not be encoded as axioms.